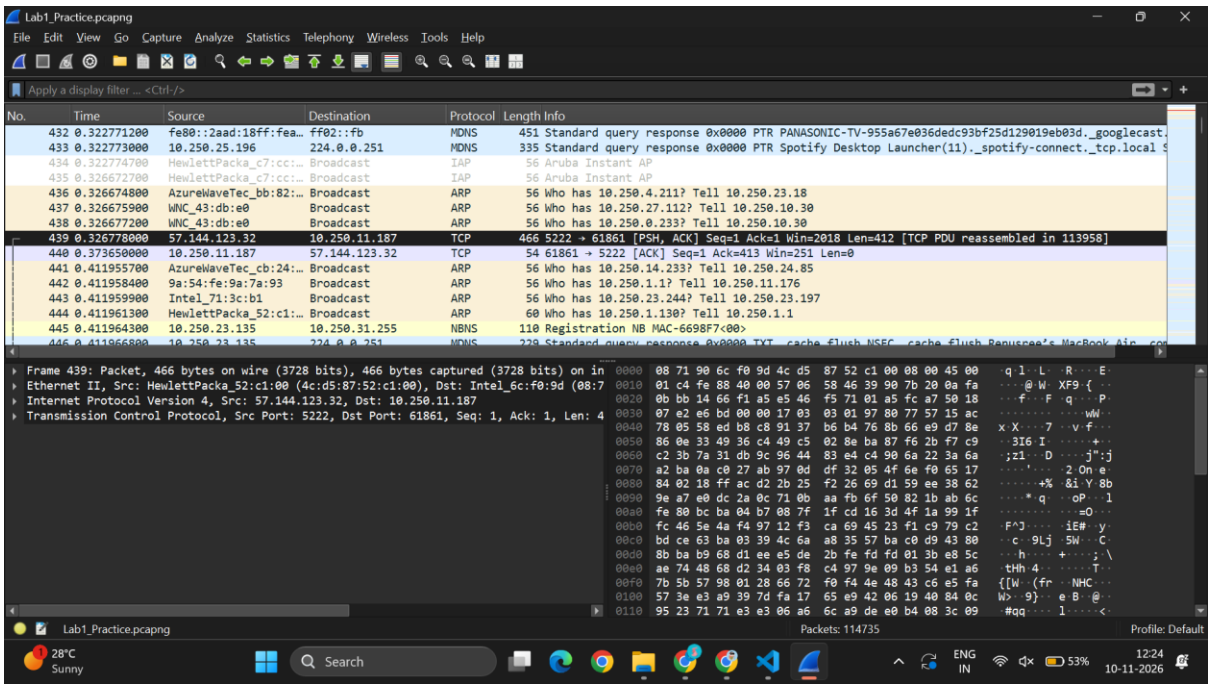
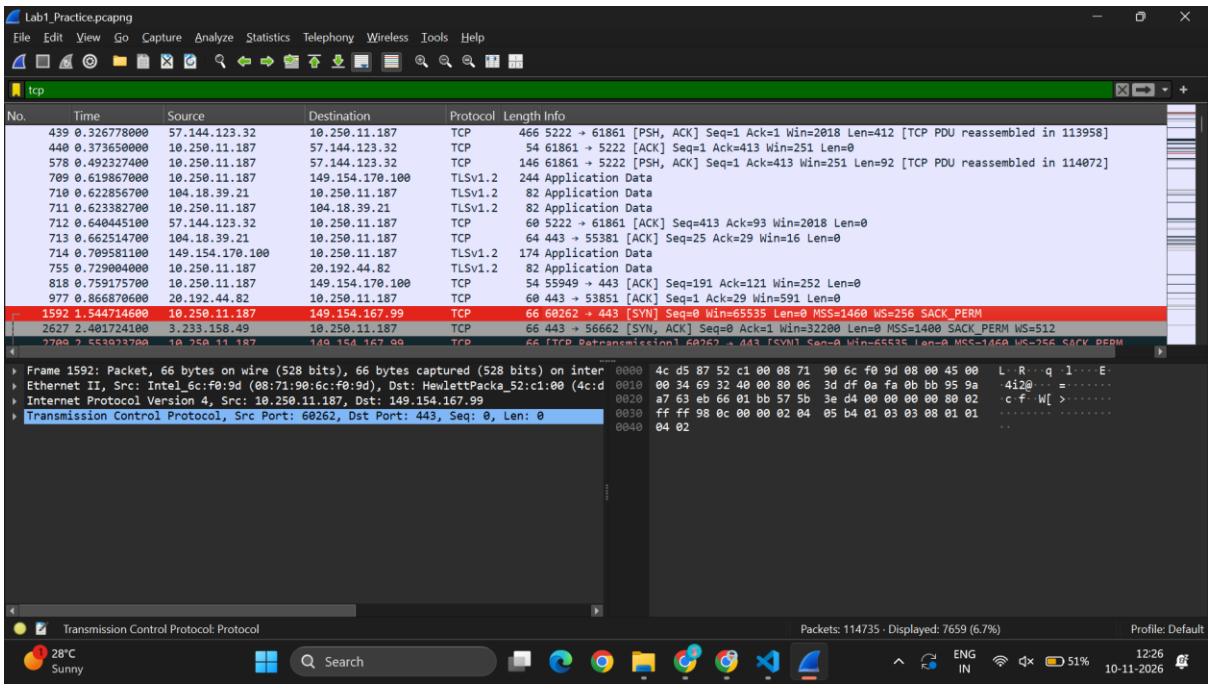
LAB-1

TASK-1

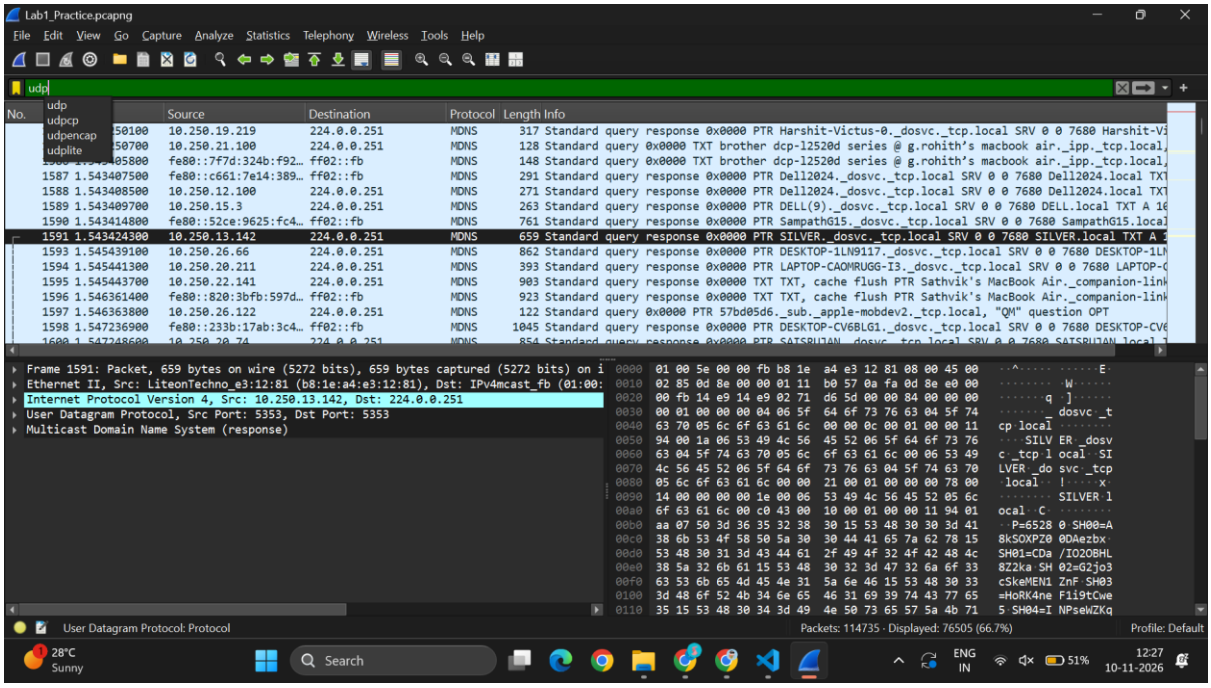
1. Open the PCAP file and identify the total number of packets captured.



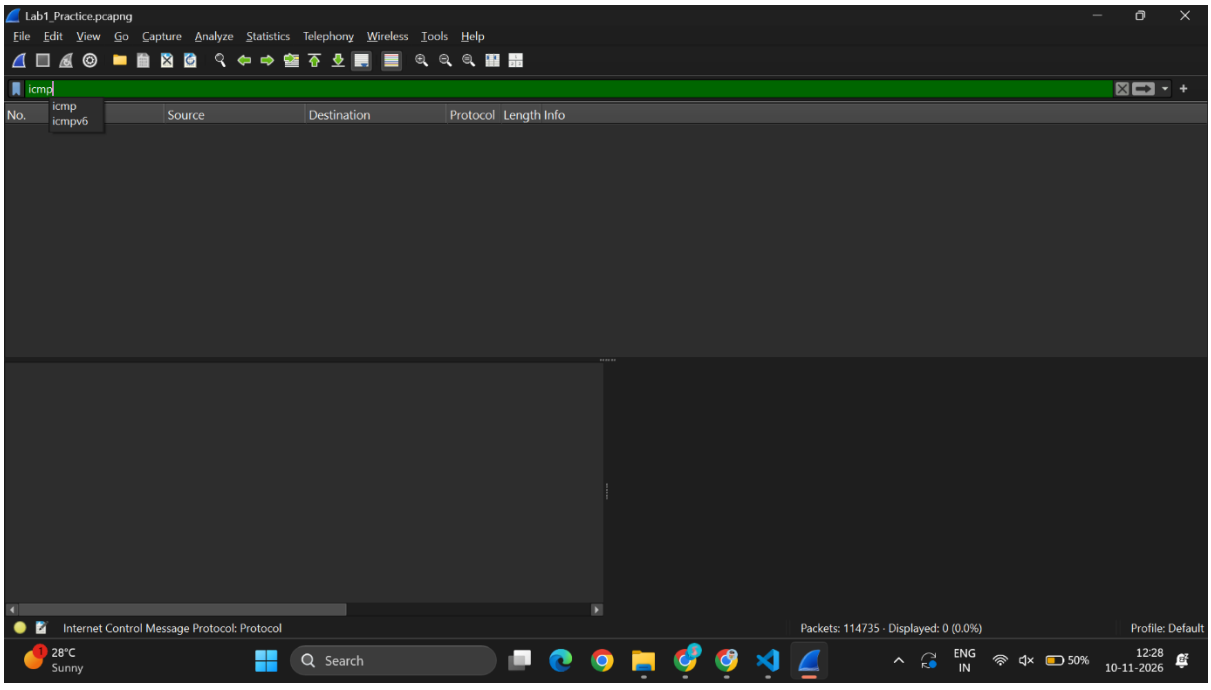
2. Apply the filter tcp and record the number of TCP packets.



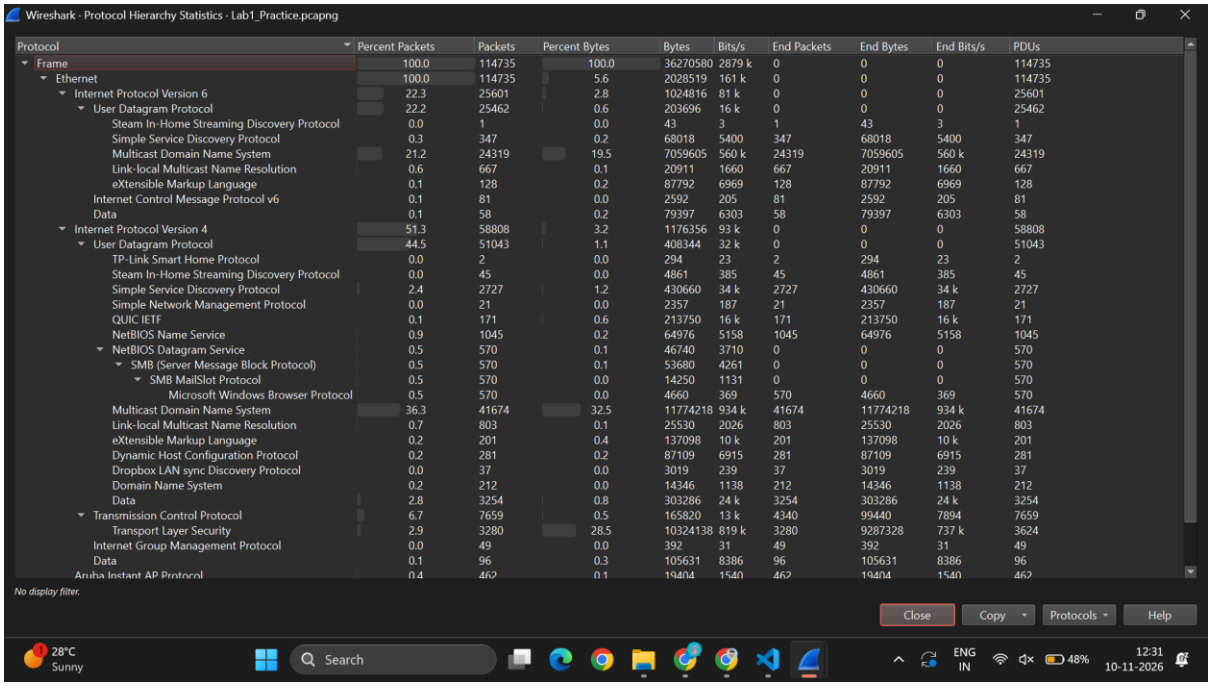
3. Apply the filter udp and record the number of UDP packets.



4. Apply the filter icmp and record the number of ICMP packets.

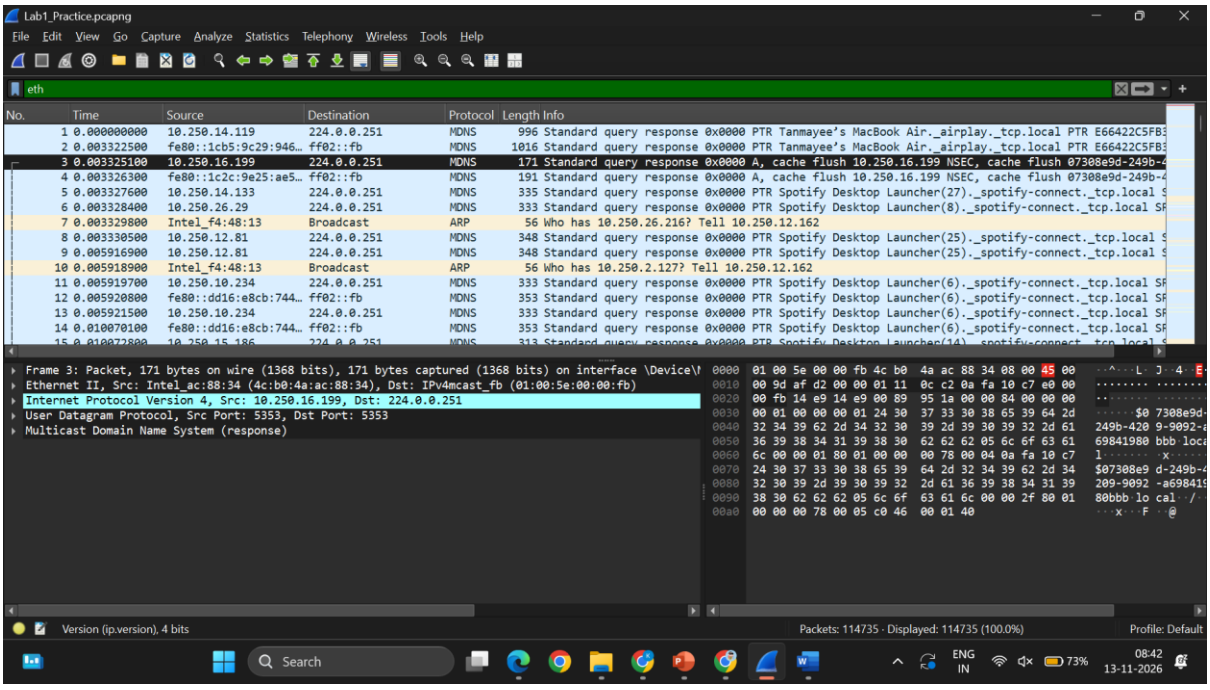


5. Identify the top five protocols using Statistics → Protocol Hierarchy.



TASK-2

6. Apply the filter eth.



7. Select five different packets and record their:
   - Frame number
   - Frame length
   - Source MAC address

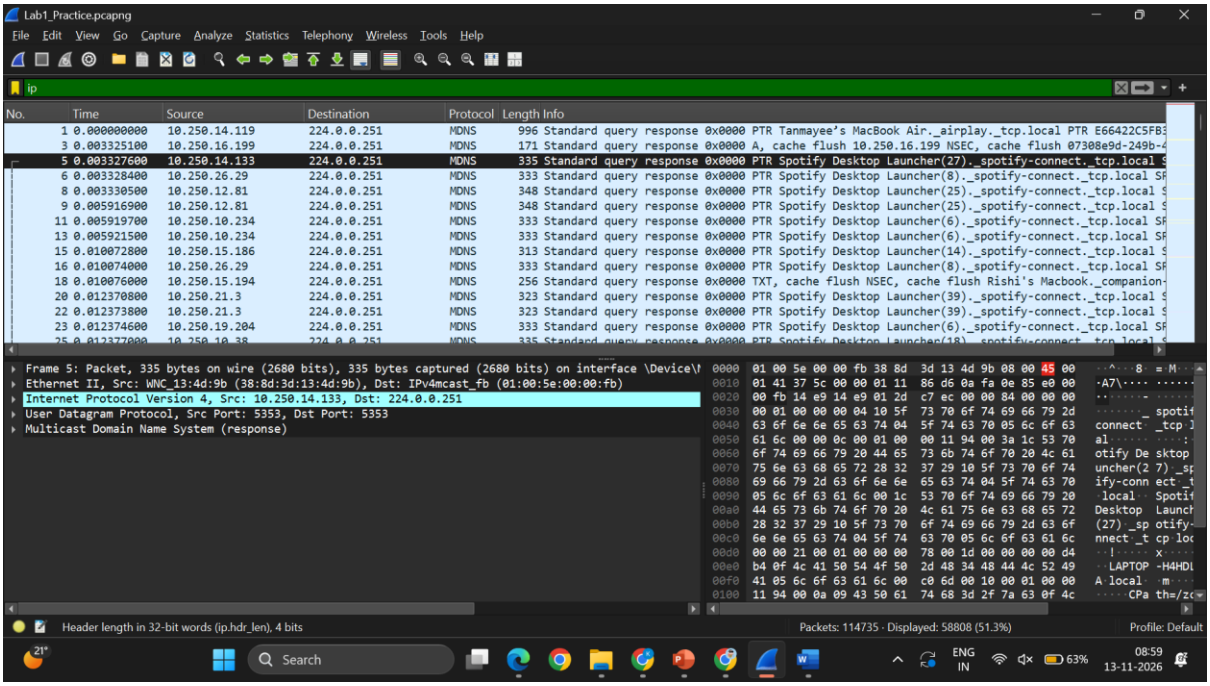
- Destination MAC address
- EtherType

packet	Frame no	Frame length	Source MAC Address	Destination MAC Address	Ether Type
1	1	996 bytes	4a:b1:f9:3d:8e:6e	01:00:5e:00:00:fb	IPv6
2	2	1016 bytes	4a:b1:f9:3d:8e:6e	33:33:00:00:00:fb	IPv6
3	3	171 bytes	4c:b0:4a:ac:88:34	01:00:5e:00:00:fb	IPv4
4	4	191 bytes	4c:b0:4a:ac:88:34	33:33:00:00:00:fb	IPv6
5	5	335 bytes	38:8d:3d:13:4d:9b	01:00:5e:00:00:fb	IPv4

8. Identify packets carrying IPv4 and IPv6 traffic.

Task 3: IPv4 packet inspection

9. Apply the filter ip.

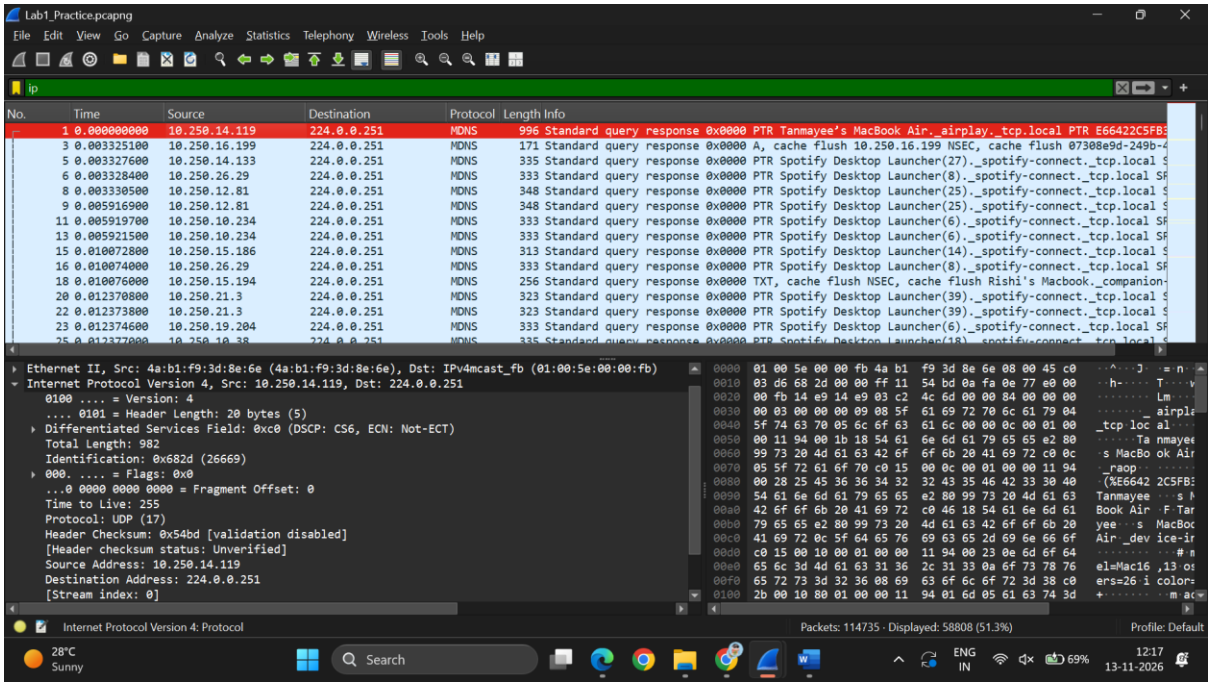


10. Identify five IPv4 packets and record:

- Source IP address
- Destination IP address
- Header length
- Total length
- TTL
- Identification field

Frame	Source IP address	Destination IP adderss	Header length	Total lenth	TTL	Identification field
1	10.250.14.119	224.0.0.251	20 bytes	982	----	26669
2	10.250.16.199	224.0.0.251	20 bytes	157	----	45010
3	10.250.14.133	224.0.0.251	20 bytes	321	----	14172
4	10.250.26.29	224.0.0.251	20 bytes	319	----	22503
5	10.250.12.81	224.0.0.251	20 bytes	334	----	7395

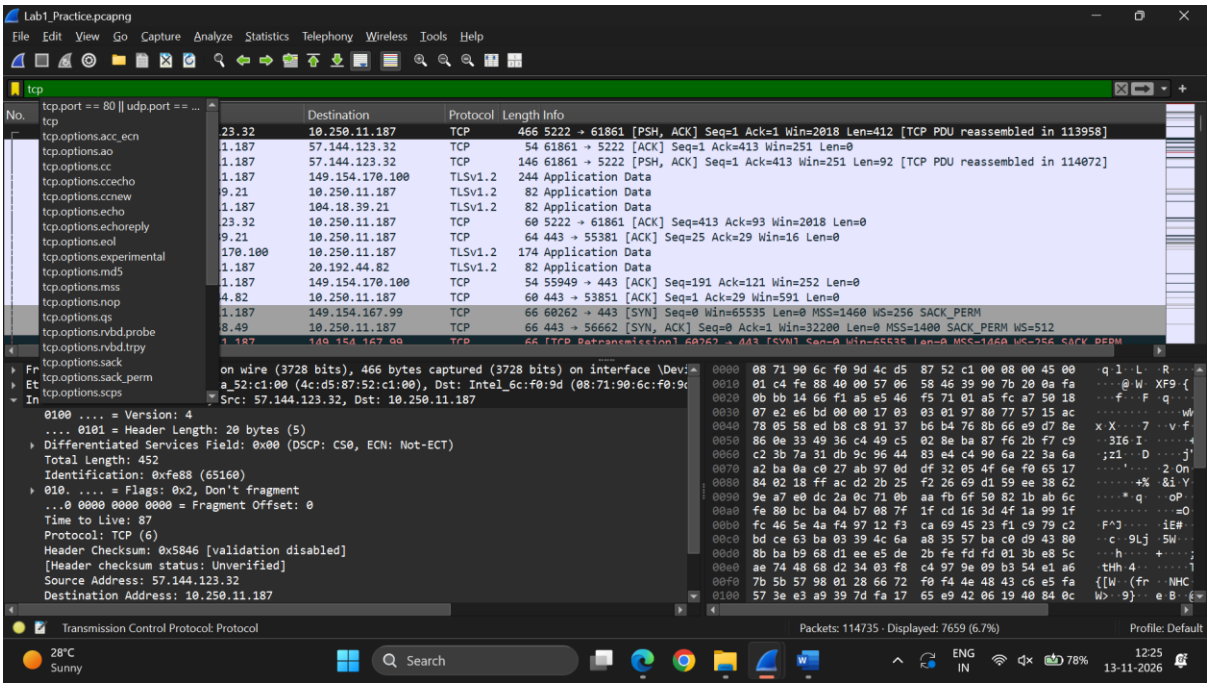
11. Find the packet with the maximum IPv4 total length.



12. Identify packets with different TTL values.

Task 4: TCP header inspection

13. Apply the filter tcp.

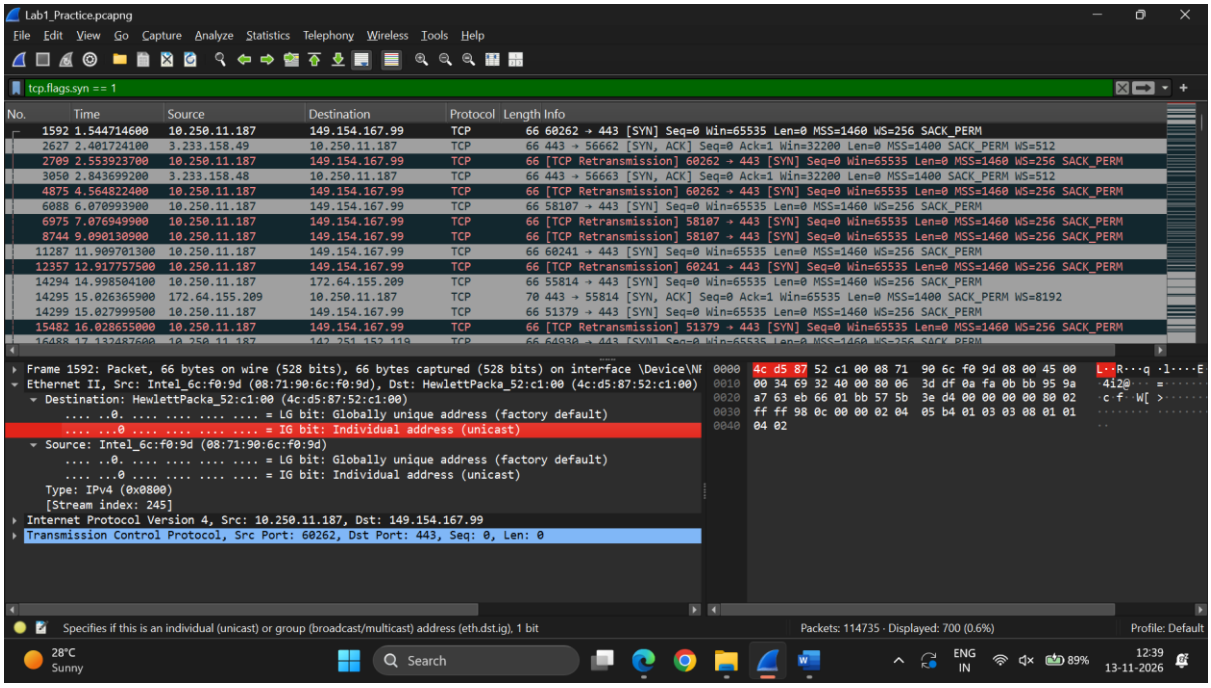


14. Select five TCP packets and record:

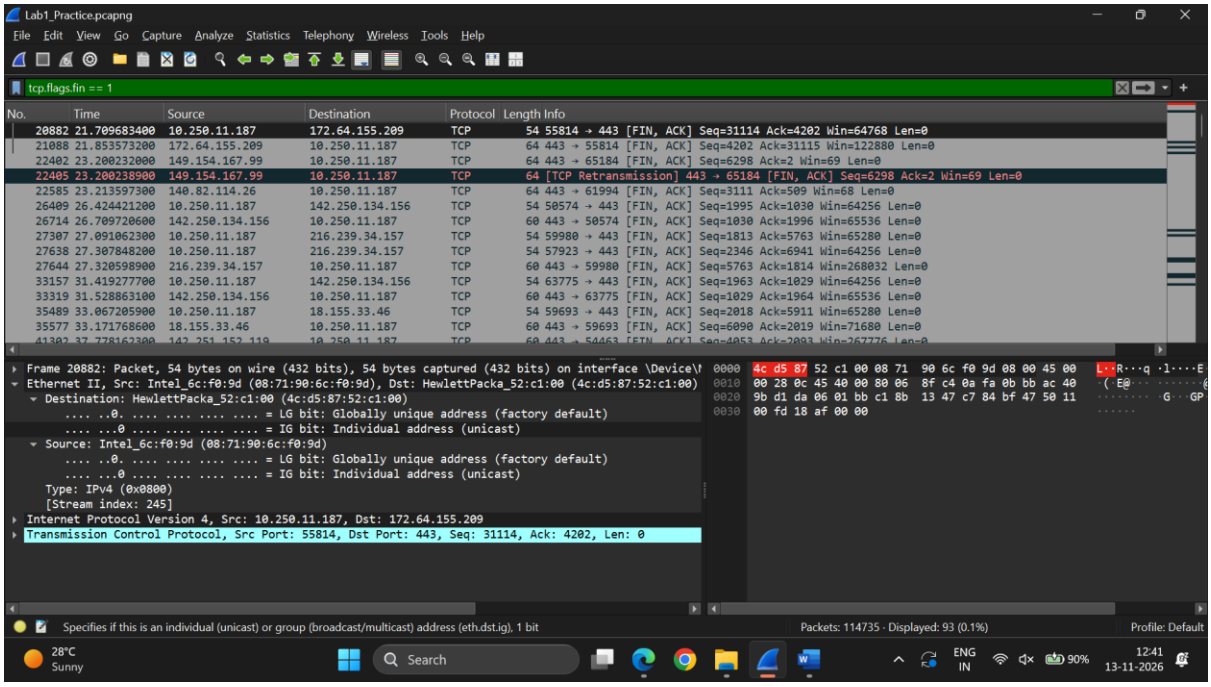
- Source port
- Destination port
- Sequence number
- Acknowledgement number
- TCP header length
- Payload length

S.No	Packet no.	TCP header length	TCP Payload Lengtth
1	439	20 bytes	412 bytes
2	440	20 bytes	0 bytes
3	578	20 bytes	92 bytes
4	709	20 bytes	190 bytes
5	710	20 bytes	24 bytes

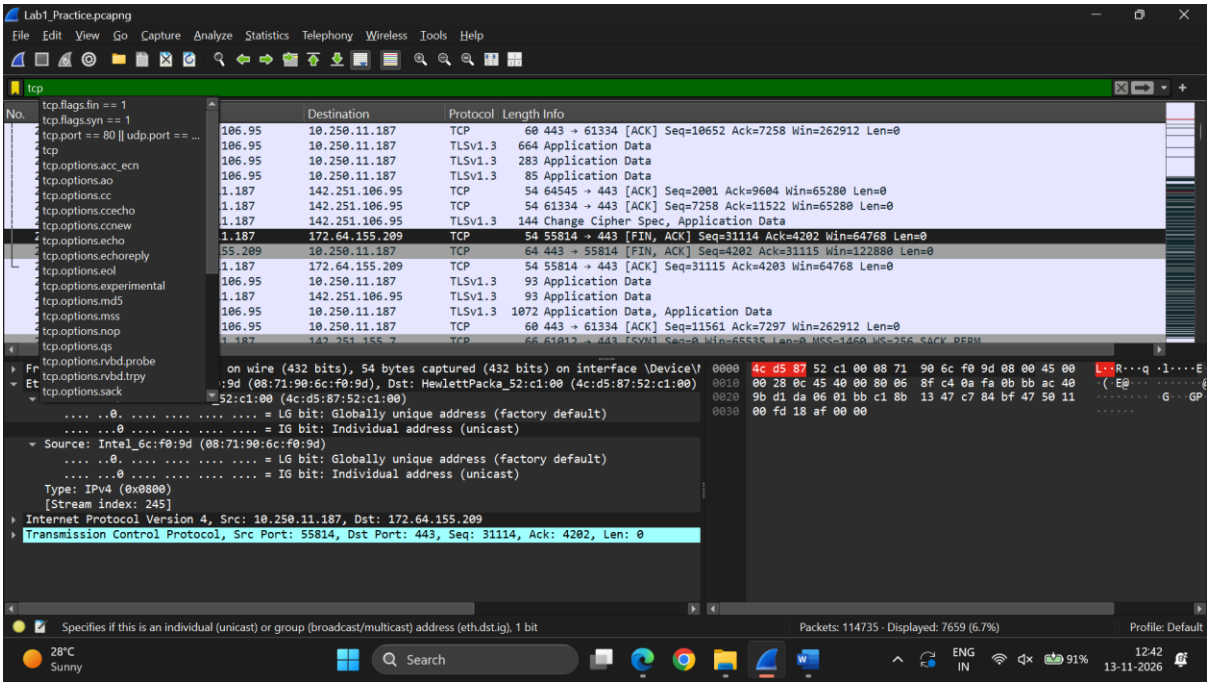
15. Apply the filter tcp.flags.syn == 1 and identify TCP connection establishment packets.



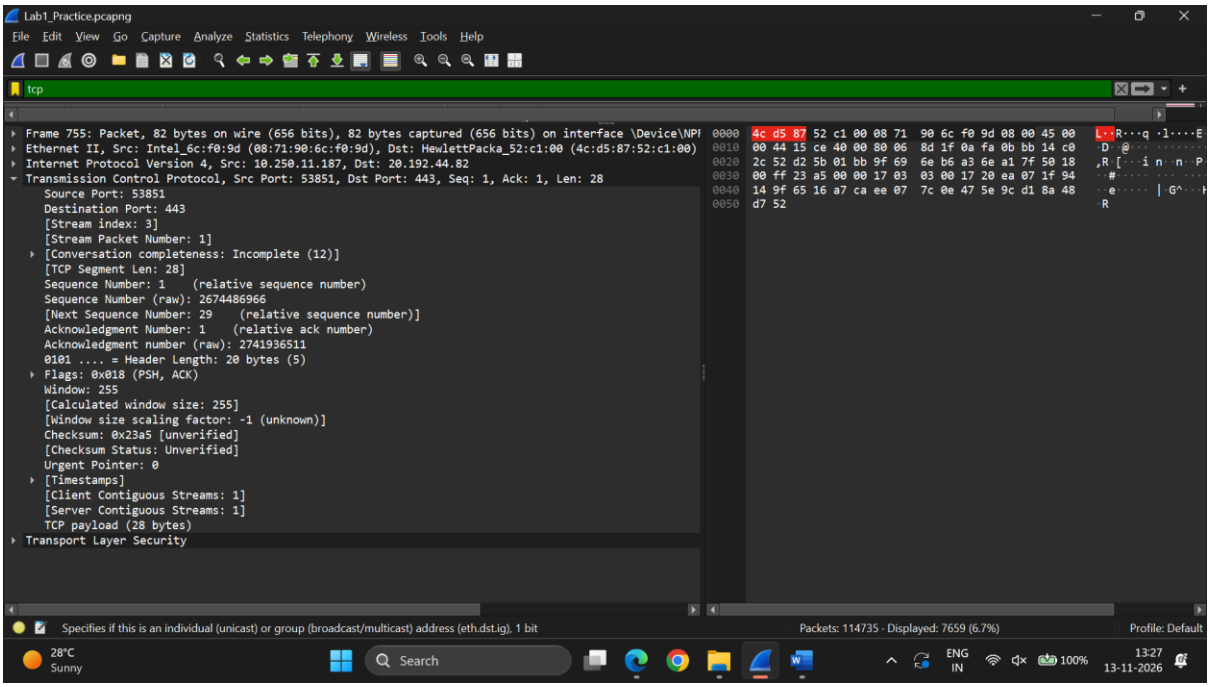
16. Apply the filter tcp.flags.fin == 1 and identify connection termination packets.



17. Identify packets containing TCP options.

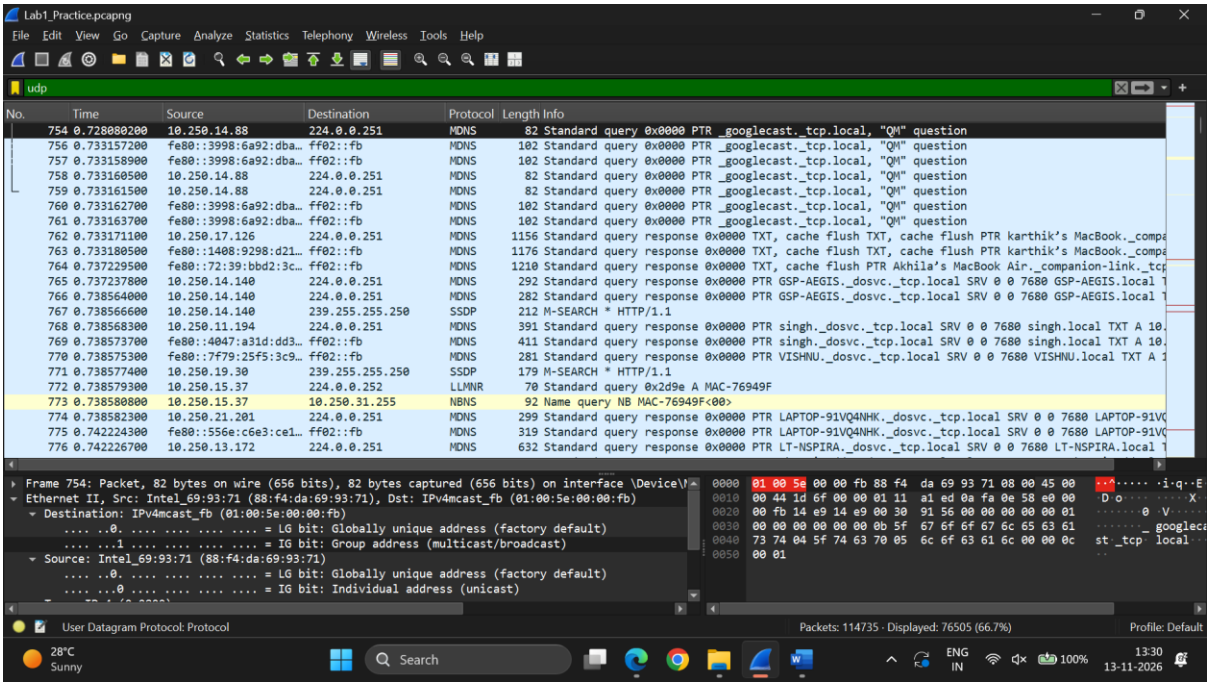


18. Find the TCP packet with the maximum payload length.



Task 5: UDP header inspection

19. Apply the filter udp.

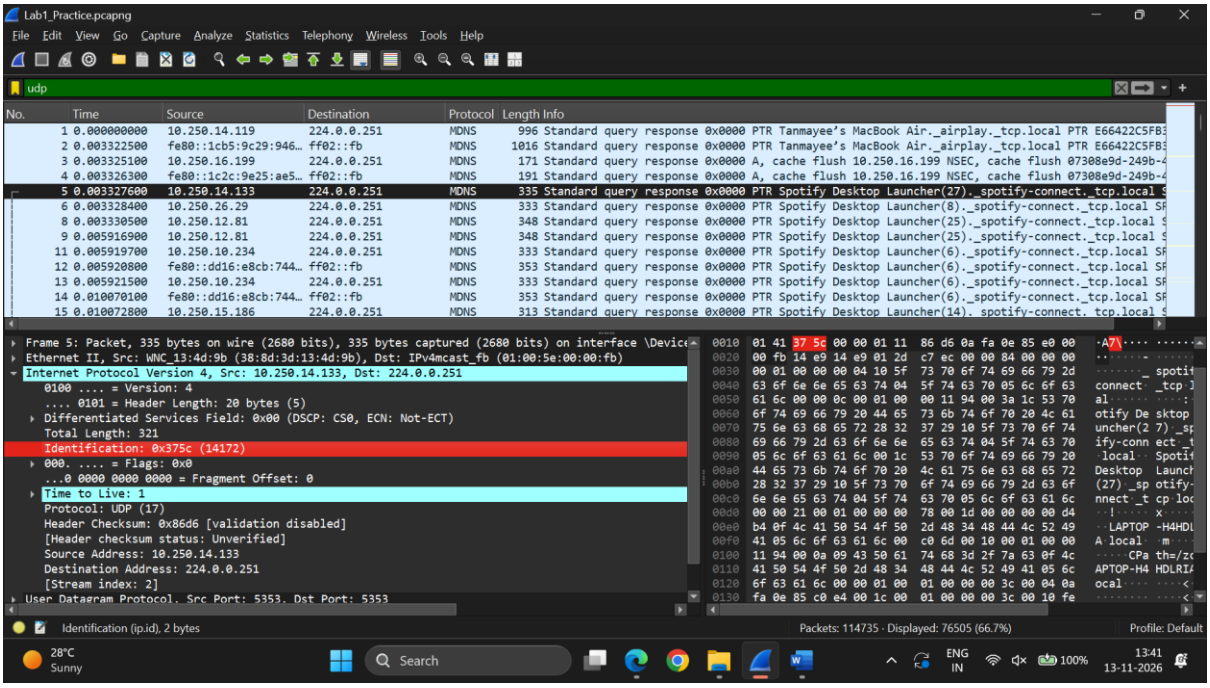


20. Select five UDP packets and record:

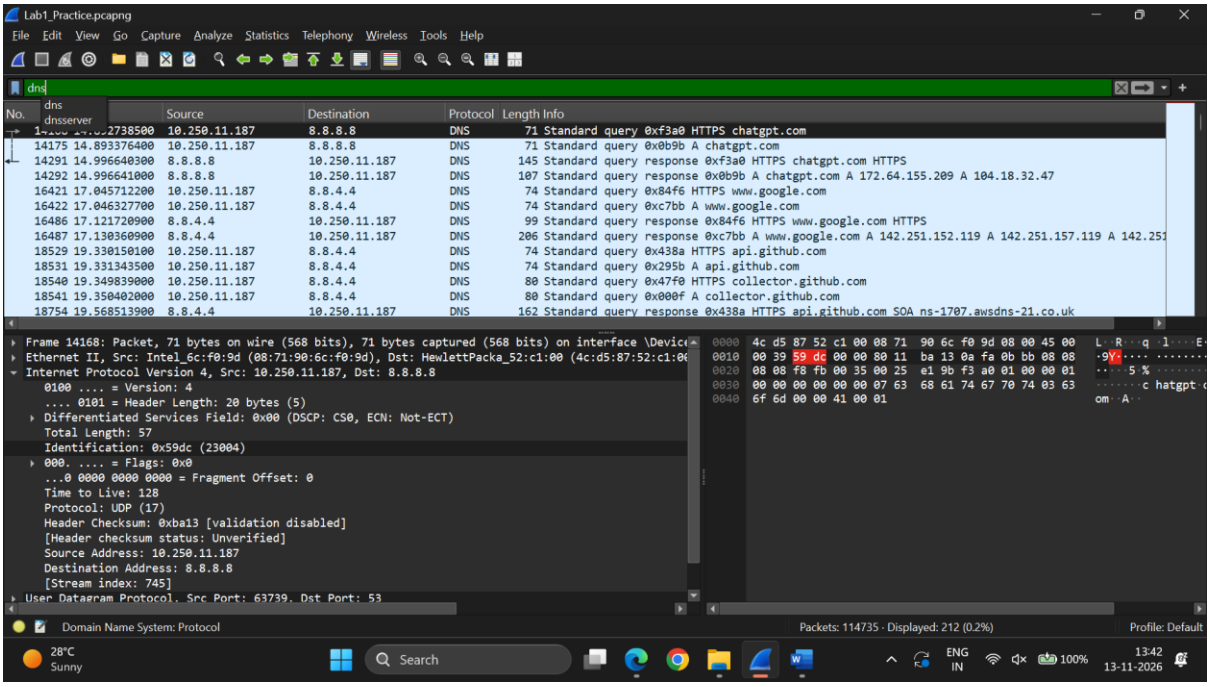
- Source port
- Destination port
- UDP length
- Payload length

s.no	Packet no	Source port	Destination port	UDP length	Payload length
1	2	5353	5353	To Check	To Check
2	5	5353	5353	To Check	To Check
3	8	5353	5353	To Check	To Check
4	12	5353	**5353**	To Check	To Check
5	14	5353	5353	To Check	To Check

21. Identify the UDP packet with the maximum payload length.

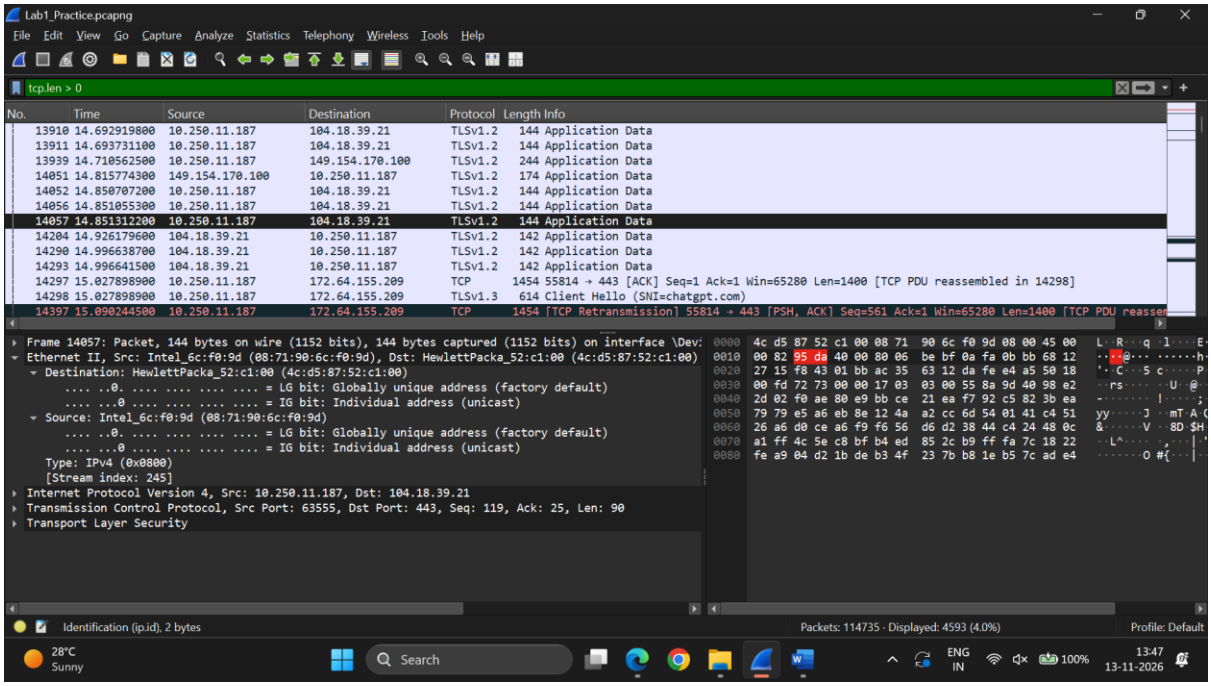


22. Find the UDP packets carrying DNS traffic using the filter dns.



Task 6: Encapsulation and overhead calculation

23. Select one TCP packet containing application data. Record the Ethernet, IPv4 and TCP header lengths.



24. Calculate the total protocol header overhead for the selected packet.

A) 54 bytes

25. Calculate the application payload size.

A)446 bytes

26. Calculate the payload efficiency percentage.

A) 89.2%

27. Repeat the same calculations for one UDP packet.

28. Compare the calculated overhead and payload efficiency of the selected TCP and UDP packets.